Mittuniversitetet
MID SWEDEN UNIVERSITY

DT081A – Signal and Image Processing

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Lecture 6
Frequency Domain

Frequency Spectrum

Fourier Transform

- The inverse Fourier Transform

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}dt$$

  - Each signal $x(t)$ is an infinite sum of sinusoids of all kinds of frequencies

$$X(\omega) = \langle x(t), e^{j\omega t} \rangle = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \in \mathbb{C} \qquad \text{(Fourier transform)}$$

- The principle of superposition
  - The sum of many sinusoids of all kinds of frequencies forms a general signal.
- A sinusoid is defined by
  - Frequency
  - Amplitude
  - Phase

Frequency Spectrum

- **Frequency Spectrum** = Fourier transform of a signal

$$X(\omega) = \langle x(t), e^{j\omega t} \rangle = F\{x(t)\} \in \mathbb{C}$$

- It has complex-values, so

$$X(\omega) = |X(\omega)| e^{j \arg X(\omega)}$$

- **Amplitude spectrum**
  - Amplitude for each frequency of the signal

$$|X(\omega)| = \sqrt{X(\omega) \cdot X^*(\omega)} = \sqrt{\left(\operatorname{Re} X(\omega)\right)^2 + \left(\operatorname{Im} X(\omega)\right)^2}$$

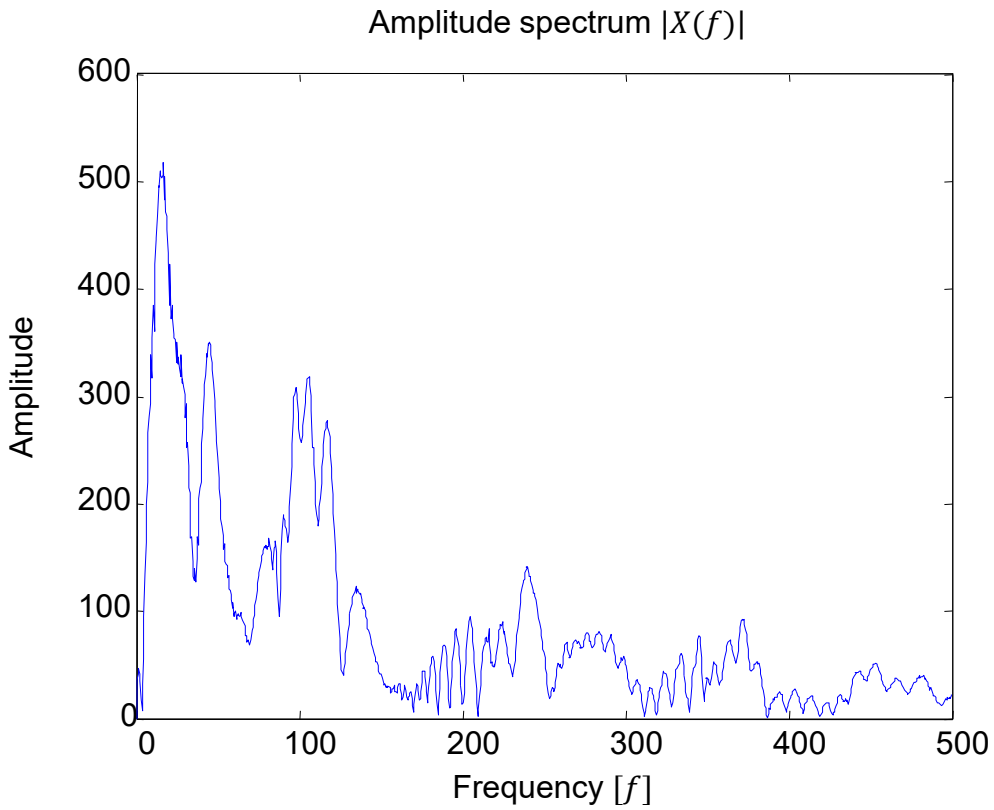
- **Phase spectrum**
  - Phase for each frequency of the signal

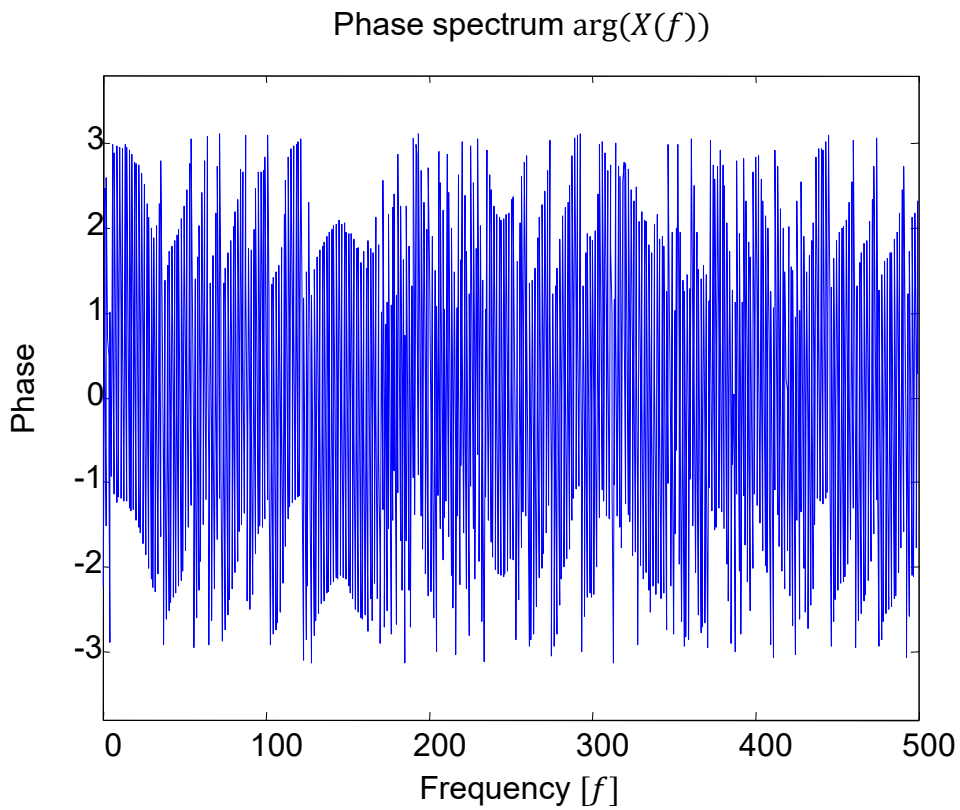
$$\arg X(\omega) = \operatorname{atan}\left(\frac{\operatorname{Im} X(\omega)}{\operatorname{Re} X(\omega)}\right) \quad (+\pi \text{ , if } \operatorname{Re} X(\omega) < 0)$$

- Frequency
- Amplitude
- Phase

Frequency Spectrum

- Frequency spectrum = the signal's frequency contents
  - I.e., what sinusoids it contains: Amplitude and Phase





System Frequency Response

Our systems

”Our” systems are

- Linear and Time-Invariant – LTI Systems $\Rightarrow$ Linear Transforms can be applied
- Often they are also *Causal*

Linear Time-Invariant Causal System

- Linear System

  Input $x = a \cdot x_1 + b \cdot x_2 \Rightarrow$ Output $y = a \cdot y_1 + b \cdot y_2$

- Time-Invariant System

  Delayed Input $x(t) = x_1(t - t_0) \Rightarrow$ Delayed Output $y(t) = y_1(t - t_0)$

- Causal system

  "Cause comes before effect"

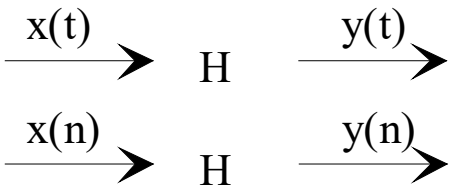
Impulse response $h(t) = \begin{cases} h^+(t), & t \geq 0 \\ 0, & t < 0 \end{cases}$



Output of an LTI System

- The output from a system is given by a convolution
  - $y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau) \cdot x(t-\tau) d\tau$ (Continuous Time System)
  - $y(n) = h(n) * x(n) = \sum_{k=-\infty}^{\infty} h(k) \cdot x(n-k)$ (Discrete Time System)

- An LTI system is fully defined by
  - Its *impulse response* $h(t)$ (Continuous Time System)
  - Its *unit pulse (impulse) response* $h(n)$ (Discrete Time System)



  - Or its transform $H(\cdot)$:
    - Fourier Transform, Laplace Transform (Continuous Time System)
    - Fourier Transform, Z Transform (Discrete Time System)

  - Or its
    - Differential equation, Ex. $y(t) = x(t) - 3 \cdot \frac{dy}{dt}$ (Continuous Time System)
    - Difference equation, Ex. $y(n) = x(n-1) - 0.5\, y(n-1)$ (Discrete Time System)

System Frequency Response (continuous time)

- The output from a system is given by a convolution
  - $y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau) \cdot x(t-\tau) d\tau$ (Continuous Time System)

- Continuous-Time Fourier Transforms
  - $X(\omega) = F\{x(t)\} \in \mathbb{C}$ (Input *Frequency spectrum*)
  - $H(\omega) = F\{h(t)\} \in \mathbb{C}$ (System *Frequency function*)
  - $Y(\omega) = F\{y(t)\} \in \mathbb{C}$ (Output *Frequency spectrum*)

- You've learnt that a convolution is transformed into a multiplication

$$y(t) = h(t) * x(t) \quad \overset{F}{\rightleftarrows} \quad \boxed{Y(\omega) = H(\omega) \cdot X(\omega)}$$

  - The **System Frequency Response** fully defines the system

System Frequency Response (discrete time)

- The output from a system is given by a convolution
  - $y(n) = h(n) * x(n) = \sum_{k=-\infty}^{\infty} h(k) \cdot x(n-k)$ (Discrete Time System)

- Continuous-Time Fourier Transforms
  - $X(e^{j\Omega}) = F\{x(\mathrm{n})\} \in \mathbb{C}$ (Input *Frequency spectrum*)
  - $H(e^{j\Omega}) = F\{h(\mathrm{n})\} \in \mathbb{C}$ (System *Frequency function*)
  - $Y(e^{j\Omega}) = F\{y(\mathrm{n})\} \in \mathbb{C}$ (Output *Frequency spectrum*)

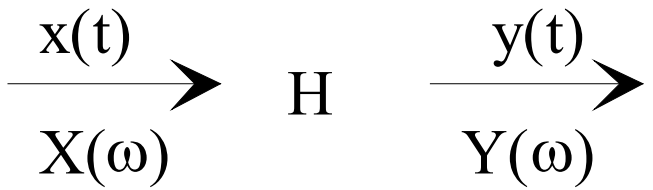
- You've learnt that a convolution is transformed into a multiplication

$$y(n) = h(n) * x(n) \quad \overset{F}{\rightleftarrows} \quad Y(e^{j\Omega}) = H(e^{j\Omega}) \cdot X(e^{j\Omega})$$

  - The **System Frequency Response** fully defines the system

Continuous Time LTI System in Frequency domain

- Continuous Time LTI System

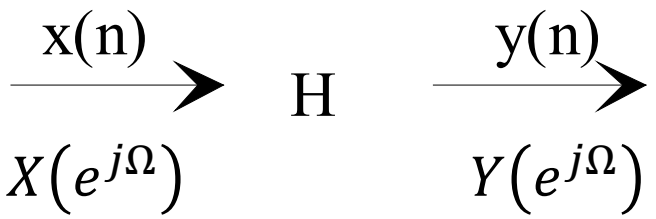


$$Y(\omega) = H(\omega) \cdot X(\omega)$$

- Frequency response $H(\omega) = |H(\omega)| \cdot e^{j \arg H(\omega)}$
- Amplification (Magnitude) $|H(\omega)|$
- Phase shift $\arg H(\omega)$
- Impulse response $h(t) = F^{-1}\{H(\omega)\}$

Discrete Time LTI System in Frequency domain

- Discrete Time LTI System



$$Y(e^{j\Omega}) = H(e^{j\Omega}) \cdot X(e^{j\Omega})$$

- Frequency response $H(e^{j\Omega}) = |H(e^{j\Omega})| \cdot e^{j \arg H(e^{j\Omega})}$
- Amplification (Magnitude) $|H(e^{j\Omega})|$
- Phase shift $\arg H(e^{j\Omega})$
- Impulse response $h(n) = F^{-1}\{H(e^{j\Omega})\}$

Consequences of the System Frequency Response

- Frequency Response $H(\omega)$
  - Decides the frequency characteristics for the LTI system

$$Y(\omega) = H(\omega) \cdot X(\omega)$$

  - Is complex, thus

$$|Y(\omega)| \cdot e^{j\arg Y(\omega)} = |H(\omega)| \cdot e^{j\arg H(\omega)} \cdot |X(\omega)| \cdot e^{j\arg X(\omega)} =$$
$$= |H(\omega)| \cdot |X(\omega)| \cdot e^{j(\arg H(\omega) + \arg X(\omega))}$$

- Amplitude
  - $|Y(\omega)| = |H(\omega)| \cdot |X(\omega)|$
  - $|H(\omega)|$ decides the amplification / damping of the input signal

- Phase
  - $\arg Y(\omega) = \arg H(\omega) + \arg X(\omega)$
  - $\arg H(\omega)$ decides the phase shift of the input signal

Consequences of the System Frequency Response

- Frequency Response $H(\omega)$
  - Decides the frequency characteristics for the LTI system

$$Y(\omega) = H(\omega) \cdot X(\omega)$$

$$|Y(\omega)| \cdot e^{j\arg Y(\omega)} = |H(\omega)| \cdot e^{j\arg H(\omega)} \cdot |X(\omega)| \cdot e^{j\arg X(\omega)}$$

- Power distribution (or Energy distribution)
  - $|Y(\omega)|^2 = |H(\omega)|^2 \cdot |X(\omega)|^2$
  - $|H(\omega)|^2$ decides how the power distribution changes by the system

Consequences of the System Frequency Response

- Output is given by input and frequency response

$$Y(\omega) = H(\omega) \cdot X(\omega)$$

- An LTI system is *frequency preserving*
  - I.e., Output contains *the same frequencies* as there is the input
  - The contribution of these frequencies may (in theory) be infinitely damped (amplitude = 0)

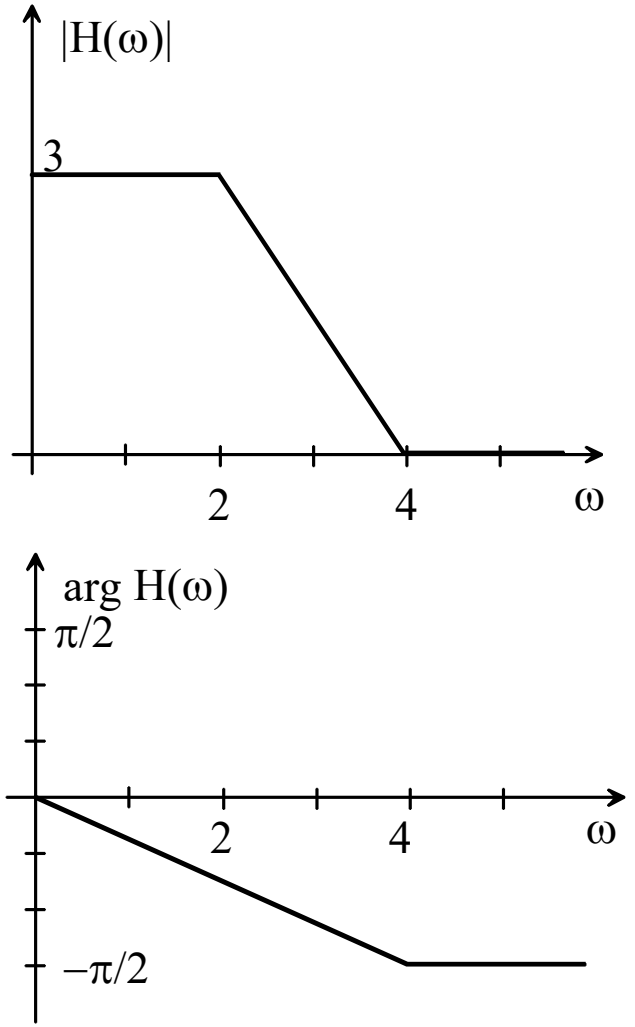
- An LTI-system affects only the input signal's
  - Amplitude
  - Phase

Example LTI system

- An LTI system has the following
  - Amplification (Magnitude) $|H(\omega)|$
  - Phase shift $\arg H(\omega)$

- Let the input be $x(t) = 2\cos(1 \cdot t + \pi/4)$
  - Amplitude $A_x = 2$
  - Phase $\phi_x = \pi/4$
  - Frequency $\omega = 1$ rad/s

- What is the output $y(t)$?

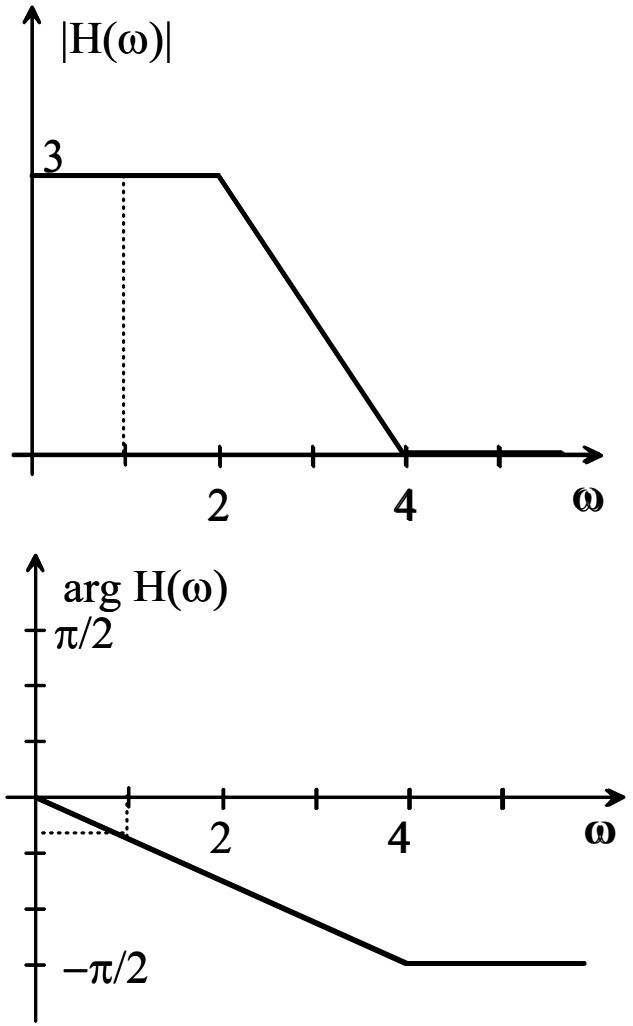


Example LTI system

- Input is $x(t) = 2\cos(1 \cdot t + \pi/4)$
- The LTI system is frequency preserving

$\Rightarrow$

- Amplitude
  - $A_y = |H(1)| \cdot |X(1)|$
    $= 3 \cdot 2 = 6$
- Phase
  - $\phi_y = \arg H(1) + \arg X(1)$
    $= -\pi/8 + \pi/4 = \pi/8$
- Therefore, the output is
  - $y(t) = 6\cos(1 \cdot t + \pi/8)$



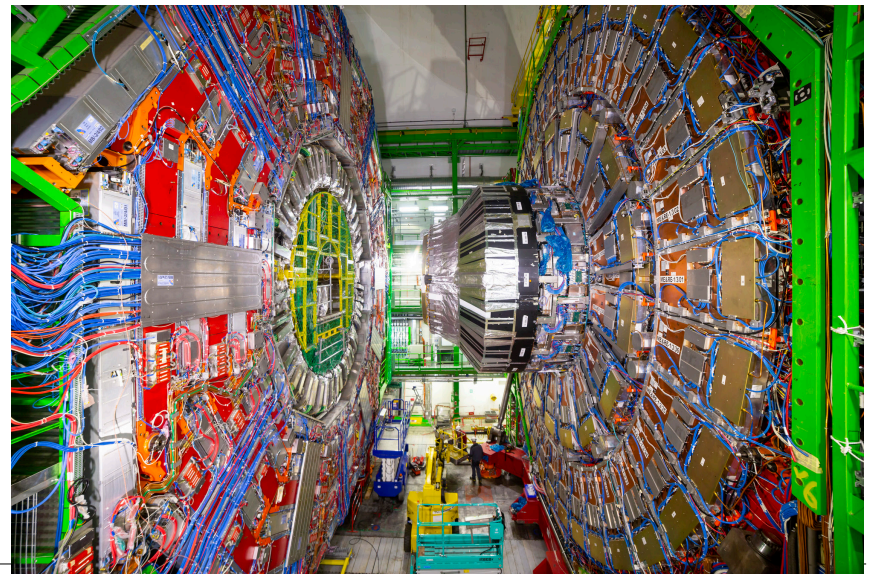
Identifying the System Frequency Response

Example CERN – Licentiate work

- Analysis Method of LTI systems
- An example from CERN







Analysis of LTI systems



- Let the system be stimulated by a known input signal $x(t)$ and measure the output $y(t)$
- The inputs signal should – in principle – contain *all frequencies* for the span one wants to investigate.
- Frequency, Amplitude and Phase of input signal $x(t)$ are known.
- Determine the output signal
  - Amplitude $A_y = |Y(\omega)|$
  - Phase $\phi_y = \arg Y(\omega)$
- Plot the diagramme

Analysis of LTI systems

- The system is stimulated by a *known* input signal:
- Many sinusoids $x_k(t) = A_k \sin(\omega_k t + \phi_k), \quad \phi_k = C_k$
- ”Chirp” signal
  - Sinusoid with increasing frequency $x(t) = A\sin(\omega(t) \cdot t + \phi), \quad \omega(t) = C_0 \cdot t$
- Impulse $x(t) = \mathrm{A} \cdot \delta(\mathrm{t}) \approx \mathrm{A} \cdot e^{-at}, t \geq 0$
- White Noise $X(\omega) = C$
  - Stochastic signal
  - Contains all frequencies (white)

Stimulation by many sinusoids

- Many sinusoids $x_k(t) = A_k \sin(\omega_k t + \phi_k), \quad \phi_k = C_k$

- Gives line spectrum
  - Must be interpolated (assume a ”nice” spectrum)
- Time consuming
  - Many measurements
- Difficult read and interpret

- (Can accelerate particles in the CERN case)

Stimulation by ”Chirp”

- Sinusoid with increasing frequency $x(t) = A\sin(\omega(t)\cdot t + \phi)\,, \quad e.g.\ \omega(t) = C_0 \cdot t$

- Input and Output signals are Fourier Transformed
  - Only one measurement
- Similar energy for all frequencies
  - Gives very good results

- (Can accelerate partices in the CERN case)

Stimulation by Impulse

- Impulse $x(t) = \mathrm{A} \cdot \delta(\mathrm{t})$
  - Approximation, e.g. $x(t) \approx \mathrm{A} \cdot e^{-at}, t \geq 0$

- Gives impulse response $h(t)$
  - That gives the Frequency response $H(\omega)$
- SNR small – narrow pulse, limited amplitude
  - Impulse must be large to give a good result
  - System can be saurated
    - Measure several times and average

- (Can accelerate particles in the CERN case)

Stimulation by White Noise

- White Noise $X(\omega) = C \ \forall \omega$

- White noise contain all frequencies
- Fourier Transform input and output signals
  - $H(\omega) = Y(\omega) \ / \ X(\omega)$
  - $X(\omega) \neq 0$!!! (it can occur!)
- Different ways to create White Noise gives differently good results
  - Many sinusoid with stochastic phase $x(t) = \sum_k \sin(\omega_k t + \phi_k), \ \ \phi_k \in \{\text{S.V.}\}$
  - ”Maximum length” Noise (ones and zeros) $x(t) = \begin{cases} 1, & p(t) = 0.5 \\ 0, & p(t) = 0.5 \end{cases}$

- (Do not accelerate particles in the CERN case)

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